Conditional Generative Modeling of Stochastic Apoptosis Kinetics via Neural Spline Flows

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Exact stochastic simulation of biochemical networks via the Gillespie Stochastic Simulation Algorithm (SSA) produces unbiased sample paths of the Chemical Master Equation, but becomes prohibitively slow when large ensembles, broad parameter sweeps, or repeated likelihood evaluations are needed. We investigate a Conditional Neural Spline Flow (CNSF) as a learned surrogate for the transition kernel of a reduced six-species Caspase-3 activation network, modelling the joint conditional distribution of molecular count increments $\Delta\mathbf{N}^{(K)}$ and elapsed time $\Delta t^{(K)}$ given the current state $\mathbf{N}_0$. The surrogate is trained on trajectories generated by the Gillespie First-Reaction Method and evaluated at three temporal resolutions: single-reaction (K=1), mesoscopic (K=100), and coarse-grained (K=300) subsampling. Across scales, the model attains test NLLs of -61.0, -31.0, and -30.7 and closely matches single-step marginals for all species and Δt Normalised Wasserstein-1 distances W_1^{norm} between CNSF and SSA single-step increments decrease for molecular species as K increases (mean $W_1^{\mathrm{norm}} \approx 0.5$–$0.7$ at K=1 vs. ≈ 0.15–0.32 at K=300), while the time coordinate remains well captured at K=1, 100 but degrades at K=300. Autoregressive rollouts from matched initial conditions show that temporal aggregation dramatically reduces mean absolute relative errors for several species and for accumulated time, but certain switch-like coordinates (pC3, I) remain challenging. Although a single neural step is not cheaper than one Gillespie reaction, a K-step surrogate update replaces K exact reactions, suggesting that subsampled CNSFs can offer effective speedups for long-horizon simulations. Our results characterise the accuracy–resolution trade-off of flow-based surrogates for stochastic biochemistry and support their use in simulation-based inference, optimal control, and high-throughput in-silico screening.

CONTENTS

I. INTRODUCTION

Apoptosis is the programmed cell-death process that occurs naturally within an organism to maintain tissue homeostasis. Its over-expression or inhibition can precipitate a diverse array of pathologies. When apoptosis is deficient, cells that should be eliminated persist, potentially leading to tumorigenesis, malignancy, or autoimmune disorders. Conversely, excessive apoptotic activity results in the deleterious loss of healthy cells, a

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phenomenon characteristic of neurodegenerative conditions such as Alzheimer's or Parkinson's disease (Bose and Chaudhari, 2019).

Traditional laboratory experimentation faces critical limitations in the study of complex biological processes such as apoptosis. Physical experiments can take hours or days and provide only fixed snapshots of the process, whereas stochastic computer simulations allow observation of the time evolution of dozens of molecular species second by second. Moreover, in a system with 40 or more chemical components, it is practically impossible to test all combinations of initial concentrations in a laboratory setting. By contrast, a virtual environment can generate thousands of scenarios to identify critical behaviors and to shortlist experiments with the highest probability of success before investing resources in expensive reagents (Raychaudhuri *et al.*, 2008).

The Gillespie Stochastic Simulation Algorithm (Gillespie, 1976a, 1977a, 2007) is the gold standard for generating exact trajectories from a chemical master equation, but its computational cost grows linearly with the number of reaction events. This cost becomes prohibitive when thousands of trajectories over wide ranges of initial conditions are required.

Recent advances in deep generative models, particularly normalizing flows (Durkan *et al.*, 2019a; Papamakarios *et al.*, 2021), suggest a complementary strategy. Instead of simulating individual reaction events, one can learn a surrogate model that directly represents the conditional transition probability of the stochastic system. Conditional normalizing flows have already been used as pseudo-reversible surrogates for SDEs (Yang *et al.*, 2024), for chemical kinetic parameter generation (Hamedi *et al.*, 2024), and, more broadly, for stochastic process emulation (Cramer *et al.*, 2024). Taken together, these studies demonstrate that an exact-likelihood generative model trained on simulation data can reproduce the full distribution of outcomes while running faster than the original simulator once training is complete.

In this work we train a Conditional Neural Spline Flow (CNSF) to emulate the one-step (or K-step) transition kernel of a reduced stochastic model of the apoptosis pathway, learning the joint distribution $P(\Delta\mathbf{N}^{(K)}, \Delta t^{(K)} \mid \mathbf{N}_0)$ for three choices of the subsampling factor $K \in \{1, 100, 300\}$.

These three values of K correspond to qualitatively distinct regimes. At $K = 1$, the target distribution is dominated by the discrete, reaction-by-reaction jumps of the SSA. At $K = 100$, the aggregated increments begin to converge toward Gaussian profiles due to a Central Limit Theorem (CLT) effect. At $K = 300$, the model must capture macroscopic, drift-dominated dynamics.

We provide a comprehensive multi-metric evaluation, negative log-likelihood (NLL), marginal histograms, Wasserstein distances, and autoregressive trajectories, to quantify the fidelity of the surrogate at each scale.

This report is organized as follows. Section II introduces the biological system and the SSA reference simulator. Section III describes the CNSF architecture and training procedure. Section IV defines the evaluation metrics. Section V presents and discusses the results for each temporal scale. Section VI summarizes the findings and outlines directions for future work.

II. BIOLOGICAL SYSTEM AND STOCHASTIC REFERENCE MODEL

A. Reduced Apoptosis Execution Network

We consider a minimal, stochastically exact model of the execution phase of apoptosis, focusing on the activation and inhibition of Caspase-3 as the main executioner protease. The network is deliberately reduced so as to retain only the dominant reactions controlling the apoptotic decision, while remaining bio-physically interpretable.

The six molecular species are S (activating stimulus, representing upstream active Caspase-8), pC3 (Pro-Caspase-3, the inactive zymogen), S·pC3 (enzyme–substrate complex), C3 (active Caspase-3), I (Inhibitor of Apoptosis Protein, representing XIAP), and C3·I (the inhibited complex). The system state at time t is $\mathbf{N}(t) = \left(N_S,\, N_{pC3},\, N_{SC3},\, N_{C3},\, N_I,\, N_{CI}\right)^\top \in \mathbb{Z}^6_{\geq 0}$.

The biochemical interactions are grouped into two elementary modules (Table I):

(R_1) S catalyses the proteolytic cleavage of pC3 via transient complex formation.

(R_2) Active Caspase-3 (C3) is reversibly sequestered by the inhibitor I.

TABLE I: Biochemical reactions and stochastic rate constants for the reduced apoptosis execution model. Second-order rate constants have units of molecule^{-1} s^{-1}; first-order rate constants have units of s^{-1}.

ID	Reaction	k^+	k^-	k_{cat}
R_{1a}	$\mathrm{S} + \mathrm{pC3} \rightleftharpoons \mathrm{S{\cdot}pC3}$	2×10^{-5}	2×10^{-4}	–
R_{1b}	$\mathrm{S{\cdot}pC3} \rightarrow \mathrm{S} + \mathrm{C3}$	–	–	2×10^{-3}
R_2	$\mathrm{C3} + \mathrm{I} \rightleftharpoons \mathrm{C3{\cdot}I}$	1×10^{-4}	1×10^{-4}	–

We declare that apoptosis has occurred when the number of active Caspase-3 molecules crosses a fixed threshold, $N_{C3} \geq \theta$ with $\theta = 12040$, which reflects the activation of a positive feedback loop that renders the cell-fate decision effectively irreversible (McIlwain *et al.*, 2013).

B. Gillespie First-Reaction Method

As a stochastic reference model, we use the First-Reaction Method (FRM) variant of the Gillespie SSA (Gillespie, 1976b, 1977b), implemented in Python with Numba JIT compilation (Lam *et al.*, 2015) for performance.

Given the current state $\mathbf{N}(t)$, the five reaction propensity functions are

$$\begin{aligned} a_0(\mathbf{N}) &= k_1^+\, N_S N_{pC3}, \\ a_1(\mathbf{N}) &= k_1^-\, N_{SC3}, \\ a_2(\mathbf{N}) &= k_{\text{cat}}\, N_{SC3}, \\ a_3(\mathbf{N}) &= k_2^+\, N_{C3} N_I, \\ a_4(\mathbf{N}) &= k_2^-\, N_{CI}. \end{aligned} \tag{1}$$

For each reaction channel $r \in \{0, \ldots, 4\}$, an independent candidate waiting time $\tau_r \sim \text{Exp}\big(a_r(\mathbf{N})\big)$ is drawn. The reaction with the smallest waiting time, $r^\star = \arg\min_r \tau_r$, is executed, the system state $\mathbf{N}$ is updated according to the corresponding stoichiometric change, and the time is advanced by $\Delta t = \tau_{r^\star}$.

The FRM is mathematically equivalent to the classical Direct Method SSA, but in this small, sparse network it requires only five exponential draws and propensity evaluations per step, making it a highly efficient and accurate stochastic reference for our surrogate modelling experiments.

C. Dataset Generation and Multi-Scale Subsampling

a. Initial conditions. To ensure that the surrogate model generalises across both physiological and pathological concentration regimes, each stochastic trajectory is initialised from discrete uniform distributions:

$$\begin{aligned} N_S &\sim U[10000,\ 20000], \\ N_{pC3} &\sim U[15000,\ 20000], \\ N_I &\sim U[2500,\ 7500], \end{aligned}$$

while the complex and active species are set to zero at $t = 0$: $N_{SC3} = N_{C3} = N_{CI} = 0$. The maximum simulation time is fixed to $T_{\max} = 3600$ s, and trajectories are terminated early if the apoptosis threshold is reached.

b. Pair formation. Each Gillespie trajectory is an ordered sequence of reaction events, $\big\{\big(\mathbf{N}(t_k),\, t_k\big)\big\}_{k\geq 0}$, where t_k is the time of the k-th event and $\mathbf{N}(t_k)$ is the corresponding state.

For a subsampling factor $\Delta \geq 1$, we construct *training pairs* by grouping every Δ consecutive reaction events:

$$(X_k,\, Y_k^{(\Delta)}) = \Big(\mathbf{N}(t_k),\, \underbrace{\mathbf{N}(t_{k+\Delta}) - \mathbf{N}(t_k)}_{\Delta\mathbf{N}_k^{(\Delta)}},\, \underbrace{t_{k+\Delta} - t_k}_{\Delta t_k^{(\Delta)}}\Big), \tag{2}$$

so that consecutive pairs within a trajectory are spaced exactly Δ reaction events apart. Here, X_k is the current state, and $Y_k^{(\Delta)} = (\Delta\mathbf{N}_k^{(\Delta)}, \Delta t_k^{(\Delta)})$ is the associated Δ-step increment in state and time.

c. Multi-scale datasets. We generated 1,000 trajectories with variable initial conditions and extracted training pairs at three temporal resolutions (subsampling factors $\Delta \in \{1, 100, 300\}$):

1. **Scale $\Delta = 1$ (single-event).** Each pair corresponds to a single elementary reaction. The increment $\Delta\mathbf{N}_k^{(1)}$ is a sparse vector with exactly one non-zero entry of ± 1.

2. **Scale $\Delta = 100$ (mesoscopic).** The increment $\Delta\mathbf{N}_k^{(100)}$ aggregates 100 consecutive reaction events. By the Central Limit Theorem, the marginal distributions of these aggregated changes begin to resemble unimodal, near-Gaussian profiles.

3. **Scale $\Delta = 300$ (coarse-grained).** The model must capture macroscopic, drift-dominated dynamics over large effective time steps. The corresponding time increments $\Delta t_k^{(300)}$ exhibit heavier-tailed distributions, reflecting variability in reaction activity.

For each scale, we perform an 80/20 split of trajectories into training and test sets (fixed random seed 42). The resulting dataset sizes are summarised in Table II.

TABLE II: Dataset statistics across the three temporal scales. N_{train} and N_{test} denote the number of training and test pairs, respectively.

Scale	Δ	N_{train}	N_{test}
Single-event	1	35693579	9010103
Mesoscopic	100	356625	89403
Coarse-grained	300	118609	29737

III. CONDITIONAL NEURAL SPLINE FLOW SURROGATE

A. Preprocessing: `ChemicalScaler`

Molecular counts in biochemical networks can span several orders of magnitude across species, whereas elapsed times Δt are typically right-skewed and approximately log-normal (as they arise from sums of exponentially distributed inter-event times). If such raw quantities are fed directly into a neural network, activation functions may saturate and different output dimensions may contribute very unevenly to the gradient during training (see, e.g., (Durkan *et al.*, 2019b)).

To address these issues, we define a `ChemicalScaler` that applies scale-appropriate transformations to each part of the data, balancing numerical ranges while preserving physically meaningful structure.

a. State $\mathbf{N}$ *(network input).* We apply *global max-scaling* to molecular counts to preserve stoichiometric proportionality:

$$\hat{x}_i = \frac{N_i}{\max_n N_i^{(n)}}, \qquad i = 1, \ldots, S, \tag{3}$$

where $\max_n N_i^{(n)}$ is the maximum observed count for species i over all training samples, and S is the number of species. This maps each species to the interval $[0, 1]$ while maintaining relative concentration ratios, which is important for mass-action kinetics.

b. Count increments $\Delta\mathbf{N}$ *(network output).* For the state increments, we use standard *z-score normalisation* on each species independently:

$$\hat{y}_{\Delta\mathbf{N},i} = \frac{\Delta N_i - \mu_{\Delta\mathbf{N},i}}{\sigma_{\Delta\mathbf{N},i}}, \qquad i = 1, \ldots, S, \tag{4}$$

where $\mu_{\Delta\mathbf{N},i}$ and $\sigma_{\Delta\mathbf{N},i}$ are the empirical mean and standard deviation of ΔN_i over the training set for species i. This centres each increment around zero and normalises its variance, helping the network treat all species on a comparable scale.

c. Elapsed time Δt *(network output).* Elapsed times are strictly positive and right-skewed, so we apply a *log–z-score* transformation:

$$\hat{y}_{\Delta t} = \frac{\log(\Delta t) - \mu_{\log \Delta t}}{\sigma_{\log \Delta t}}, \tag{5}$$

where $\mu_{\log \Delta t}$ and $\sigma_{\log \Delta t}$ are the empirical mean and standard deviation of $\log(\Delta t)$ over the training set. The corresponding inverse transform,

$$\Delta t = \exp\bigl(\hat{y}_{\Delta t}\, \sigma_{\log \Delta t} + \mu_{\log \Delta t}\bigr), \tag{6}$$

ensures that sampled time steps remain strictly positive. This transform approximately linearises the exponential tail of the inter-event time distribution and helps balance the log-likelihood contributions from Δt and $\Delta\mathbf{N}$ during training.

d. Combined target vector. After preprocessing, the full target vector passed to the flow model is $\mathbf{y} = (\hat{y}_{\Delta\mathbf{N},1}, \ldots, \hat{y}_{\Delta\mathbf{N},S}, \hat{y}_{\Delta t}) \in \mathbb{R}^{S+1}$, with $S = 6$ species in our system.

B. Architecture: Encoder and Conditional NSF

The CNSF decomposes the learning problem into two components (Fig. 1): an *encoder* that maps the scaled molecular state to a low-dimensional context vector, and a *conditional normalising flow* that models the distribution of increments conditioned on this context.

a. Encoder Enc_ψ. The encoder is a three-layer feed-forward network with LayerNorm and SiLU activations:

$$\begin{aligned}\mathbf{h} &= \mathrm{Enc}_\psi(\hat{\mathbf{x}}) \\ &= \bigl(\mathrm{LayerNorm} \circ \mathrm{SiLU} \circ \mathrm{Linear}\bigr)^3(\hat{\mathbf{x}}) \in \mathbb{R}^H,\end{aligned} \tag{7}$$

with hidden dimension $H = 128$. LayerNorm stabilises activations across the range of scaled inputs and improves gradient flow, which is particularly important given the near-zero initial state at $t = 0$.

b. Conditional NSF. The conditional flow $F_\phi : \mathbb{R}^{S+1} \to \mathbb{R}^{S+1}$ is constructed as a composition of $L = 6$ coupling layers. Each layer consists of:

1. a random permutation of the dimensions; and
2. a Piecewise Rational Quadratic Spline (RQS) coupling transform (Durkan *et al.*, 2019b).

In a given coupling layer, the input vector is partitioned into *passive* and *active* subsets according to a fixed mask (e.g. $\mathbf{m} = [1, 0, 1, 0, \ldots]$). The passive dimensions are left unchanged, while the active dimensions are updated by an invertible, context-conditioned spline transform. Formally, for layer l,

$$\begin{aligned}\mathbf{z}_{\mathrm{pass}}^{(l)} &= \mathbf{z}_{\mathrm{pass}}^{(l-1)}, \\ \mathbf{z}_{\mathrm{act}}^{(l)} &= \mathrm{RQS}\Bigl(\mathbf{z}_{\mathrm{act}}^{(l-1)}\ ;\ \xi\bigl(\mathbf{z}_{\mathrm{pass}}^{(l-1)}, \mathbf{h}\bigr)\Bigr),\end{aligned} \tag{8}$$

where $\xi(\cdot, \cdot)$ denotes the spline parameters.

The spline parameters $\xi(\mathbf{z}_{\mathrm{pass}}, \mathbf{h})$ are produced by a 2-block ResidualNet (He *et al.*, 2016) with 128 hidden units, conditioned on the concatenation of the passive dimensions and the context vector $\mathbf{h}$. Each RQS transform uses $B = 16$ monotone bins on the domain $[-5, 5]$ with linear tails outside this range, providing $O(B)$ flexibility while preserving analytic invertibility.

Random permutations between coupling layers ensure that all dimensions are eventually transformed.

c. Exact log-likelihood. Because the RQS coupling has an analytic Jacobian determinant, the conditional log-likelihood of a preprocessed target vector $\mathbf{y}$ given a scaled input state $\hat{\mathbf{x}}$ is

$$\log p_\theta(\mathbf{y} \mid \hat{\mathbf{x}}) = \log p_Z\Big(F_\phi^{-1}\big(\mathbf{y};\, \mathrm{Enc}_\psi(\hat{\mathbf{x}})\big)\Big) + \sum_{l=1}^{L} \log\Big|\det J_{F_l^{-1}}(\mathbf{y}_l;\, \mathbf{h})\Big|, \tag{9}$$

where F_l^{-1} is the inverse of the l-th coupling layer, $\mathbf{y}_l$ is its input, and $p_Z = \mathcal{N}(\mathbf{0}, \mathbf{I}_{S+1})$ is the standard normal base density in $\mathbb{R}^{S+1}$.

The likelihood is computed exactly and differentiably, enabling gradient-based training and calibrated uncertainty estimates.

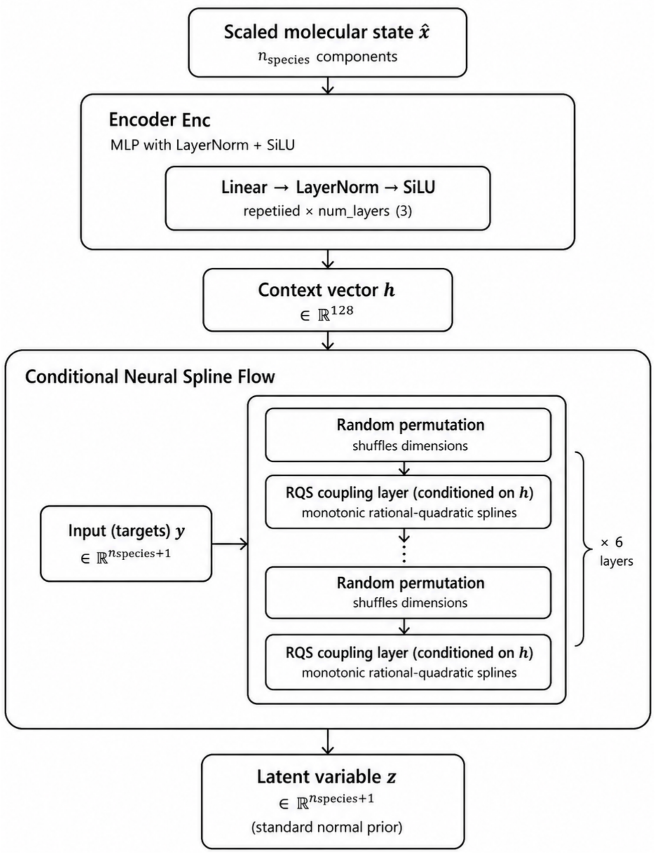


FIG. 1: Architecture of the Conditional Neural Spline Flow. The encoder Enc_ψ maps the scaled molecular state $\hat{\mathbf{x}}$ to a context vector $\mathbf{h} \in \mathbb{R}^{128}$. A stack of six RQS coupling layers then transforms the target vector $\mathbf{y}$ to the standard normal latent space $\mathbf{z}$, with each coupling network conditioned on $\mathbf{h}$. Random permutations between layers ensure that all dimensions are eventually updated.

d. Autoregressive sampling. At inference time, the CNSF is used as an autoregressive transition model. Given the current state $\mathbf{N}_k$ and its scaled version $\hat{\mathbf{x}}_k$, we:

1. compute the context $\mathbf{h}_k = \mathrm{Enc}_\psi(\hat{\mathbf{x}}_k)$;
2. draw a latent sample $\mathbf{z}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$;
3. transform it through the flow to obtain $\mathbf{y}_k = F_\phi(\mathbf{z}_k;\, \mathbf{h}_k)$;
4. invert the preprocessing to recover $(\Delta\mathbf{N}_k, \Delta t_k)$.

The next state and time are then updated as

$$\begin{aligned} \mathbf{N}_{k+1} &= \mathrm{clip}\big(\mathrm{round}(\mathbf{N}_k + \Delta\mathbf{N}_k),\, 0,\, \infty\big), \\ t_{k+1} &= t_k + \Delta t_k, \end{aligned} \tag{10}$$

where the round-and-clip operation enforces non-negative integer molecular counts.

C. Training Procedure

The model is trained to minimise the mean negative log-likelihood over training pairs $(\hat{\mathbf{x}}_n, \mathbf{y}_n)$:

$$\mathcal{L}_{\mathrm{NLL}}(\theta) = -\frac{1}{N} \sum_{n=1}^{N} \log p_\theta(\mathbf{y}_n \mid \hat{\mathbf{x}}_n). \tag{11}$$

Table III summarises the training configuration.

TABLE III: Training hyperparameters, common across all three temporal scales. Batch sizes correspond to $k = 1$, $k = 100$, $k = 300$ respectively

Hyperparameter	Value
Optimizer	Adam, β_1=0.9, β_2=0.999
Learning rate	10^{-3}
Batch size	16384, 2048, 1024
Grad. clipping	ℓ_2 norm ≤ 1.0
LR schedule	ReduceLROnPlateau patience 20, factor 0.5
Early stopping	Patience 50, test NLL
Max epochs	500
Enc. layers/width	3 / 128 (LayerNorm+SiLU)
Flow coupling layers	6
ResNet hidden/blocks	128 / 2
Spline bins/tail	16 / 5.0

The model with the lowest test NLL is retained (best-checkpoint strategy). All experiments were conducted on a single GPU using PyTorch (Paszke *et al.*, 2019) and the `nflows` library (Durkan *et al.*, 2020).

IV. EVALUATION FRAMEWORK

We design a four-tier evaluation protocol that probes the surrogate at increasing levels of temporal aggregation and trajectory length.

A. Negative Log-Likelihood (NLL)

The test-set NLL provides a single, calibrated scalar measure of how well the learned density p_θ approximates the true conditional transition kernel. A lower NLL indicates a closer approximation to the SSA distribution. Because NLL is scale-sensitive, values across different subsampling factors are not directly comparable; they serve as within-scale diagnostics.

B. Single-Step Marginal Histograms

For each of the $S = 6$ species and for Δt, we compare the empirical marginal distributions of the SSA test set against 10000 samples drawn from the CNSF conditioned on the same test inputs. Histograms are displayed in original (unscaled) units: molecule counts for $\Delta\mathbf{N}$ and seconds for Δt (with Δt plotted on a $\log_{10}$ axis). Visual inspection captures features that aggregate metrics may miss: zero inflation, multi-modality, and tail behaviour.

C. Normalised Wasserstein Distance Analysis

The Wasserstein-1 distance $W_1(P, Q)$ (Villani, 2009) is a statistically principled measure of distributional discrepancy that is sensitive to both the shape and location of the distributions. For comparability across species with vastly different concentration scales, we normalise by the standard deviation of the reference distribution:

$$W_1^{\text{norm}}(P, Q) = \frac{W_1(P, Q)}{\text{std}(P)}. \tag{12}$$

For each of 100 initial states $\mathbf{N}_0^{(j)}$ drawn uniformly at random from the test set, we generate $M = 10000$ one-step samples from both the SSA and the CNSF, and compute W_1^{norm} independently for each species and for $\log_{10}(\Delta t)$. The resulting distributions of W_1^{norm} across the 100 initial states are displayed as species-level boxplots and a time-metric histogram. The analysis identifies the initial condition with the lowest total W_1^{norm} as the "best" reference for trajectory comparison.

D. Autoregressive Trajectory Evaluation

Starting from the best initial condition identified in Sect. IV.C, we generate a single reference SSA trajectory, subsample it at the appropriate Δ factor, and compare it to a CNSF trajectory of the same number of predicted steps. The CNSF trajectory is generated autoregressively via Eq. (10).

For each species i and timestep index k, the per-step relative error is

$$\varepsilon_{k,i} = \frac{\hat{N}_{k,i} - N_{k,i}}{\max(|N_{k,i}|, 1)}, \tag{13}$$

and the mean absolute relative error over all K steps is

$$\text{MARE}_i = \frac{1}{K} \sum_{k=1}^{K} |\varepsilon_{k,i}|. \tag{14}$$

Accumulated time is evaluated analogously, with the relative error on t_k^{NN} vs. $t_k^{\text{Gillespie}}$.

V. RESULTS AND DISCUSSION

A. Scale $\Delta = 1$: Single-Event Subsampling

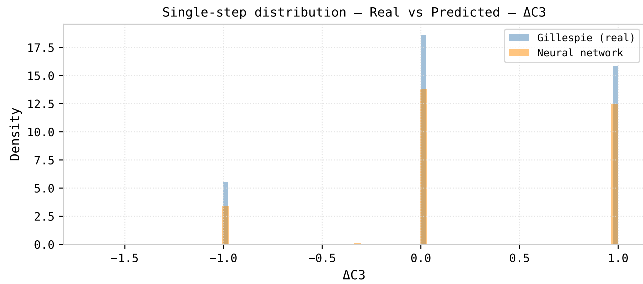
a. Negative Log-Likelihood. At single-event resolution, the target distribution $P(\Delta\mathbf{N}^{(1)}, \Delta t^{(1)} \mid \mathbf{N}_0)$ is maximally structured: exactly one reaction fires per step, so $\Delta\mathbf{N}^{(1)}$ is a sparse vector with a single non-zero entry of ± 1 among the six species. The elapsed time $\Delta t^{(1)}$ follows a mixture of truncated exponentials determined by the five propensity functions. The test NLL for this scale is -61.0097 (Table IV).

TABLE IV: Test-set Negative Log-Likelihood for each temporal scale. Lower values indicate closer agreement with the SSA transition kernel.

Scale	Δ	NLL (test)
Single-event	1	-61.0097
Mesoscopic	100	-31.0363
Coarse-grained	300	-30.6531

b. Single-Step Histograms. Figure 2 shows the marginal histograms for the Caspase-3 increment ΔN_{C3} and the elapsed time Δt at $\Delta = 1$. The spike-and-mass structure of the ΔN_{C3} histogram, with almost all probability mass concentrated at $\{-1, 0, +1\}$, poses a non-trivial challenge for a continuous density model: the flow must concentrate most of its density on three very narrow bins while suppressing density elsewhere. The elapsed-time histogram, plotted on a $\log_{10}$ scale, tests whether the model correctly captures the exponential decay characteristic of Markovian reaction times.

c. Wasserstein Analysis. As explained in Section IV.C, we compute normalised Wasserstein distances W_1^{norm} for 100 initial conditions drawn from the test set, using $M = 10000$ one-step samples per method and per initial state. Figure 3 shows the resulting discrepancies at $\Delta = 1$: for



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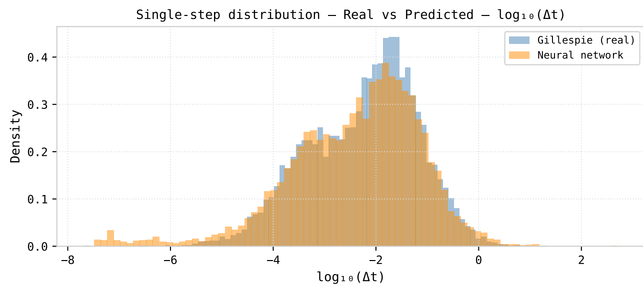


(b)

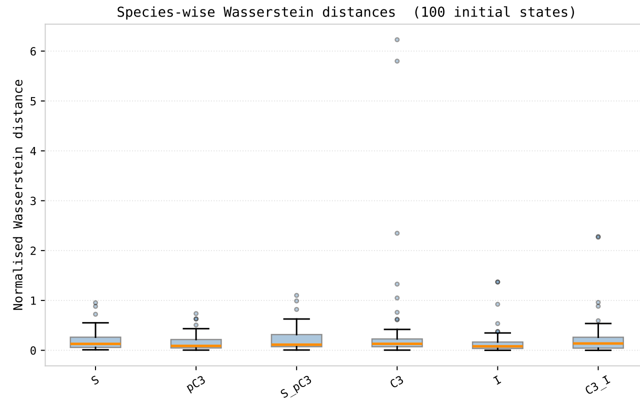
FIG. 2: Single-step marginal distributions at scale $\Delta = 1$ for the Caspase-3 increment and the elapsed time. Each panel overlays the empirical SSA histogram (blue) against 10000 CNSF samples (orange). The Caspase-3 increment ΔN_{C3} is shown on a linear axis; the elapsed time Δt is shown on a $\log_{10}$ axis.

each species, the 100 values of W_1^{norm} for the single-step increments ΔN_i are summarised as boxplots.

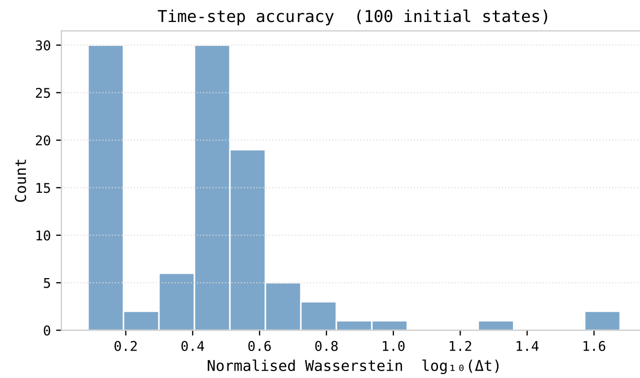
The time panel in Figure 3 displays the histogram of the W_1^{norm} values associated specifically with the time coordinate, i.e. the normalised Wasserstein distances computed on the $\log_{10}(\Delta t)$ distributions for the same 100 initial conditions. Summary statistics (mean, standard deviation, minimum, maximum of W_1^{norm}) are reported in Table V.

Figure 4 illustrates the single-step distributions of ΔN_{C3} and Δt for the *best* initial condition, i.e. the one with the lowest total W_1^{norm}.

d. Autoregressive Trajectories. Figure 5 compares SSA and CNSF trajectories for Caspase-3 and for the accumulated time at $\Delta = 1$, starting from the best initial condition identified by the Wasserstein analysis. Panels 5a and 5c show the corresponding trajectories, while Panels 5b and 5d show the associated relative error profiles (Eq. (13)). In both coordinates, the autoregressive propagation of small per-step errors causes the CNSF trajectories to diverge from the SSA reference before the end of the simulation horizon, illustrating the cumulative



(a)



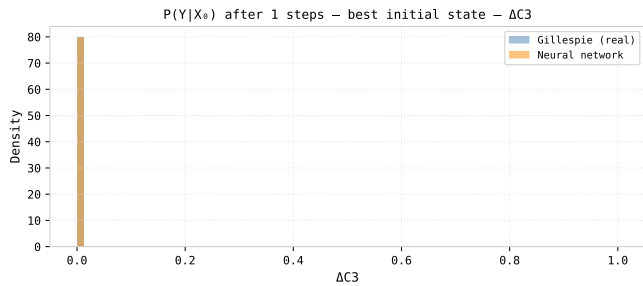
(b)

FIG. 3: Normalised Wasserstein-1 distances W_1^{norm} across the 100 initial conditions from Section IV.C at scale $\Delta = 1$. *Top:* Boxplots of W_1^{norm} for the single-step increments ΔN_i of each species; boxes span the interquartile range, whiskers extend to $1.5\times$ IQR. *Bottom:* Histogram of the 100 W_1^{norm} values for the time dimension, i.e. for the $\log_{10}(\Delta t)$ distributions.

impact of transition errors.

B. Scale $\Delta = 100$: Mesoscopic Temporal Aggregation

a. Negative Log-Likelihood. Aggregating 100 consecutive reactions transforms the target from the sparse, discrete distribution of the single-event scale to a smoother, multivariate profile that is more amenable to continuous density estimation. By the CLT, the marginals of $\Delta \mathbf{N}^{(100)}$ approach Gaussian shapes for sufficiently large ensembles, and the zero-inflation characteristic of $\Delta \mathbf{N}^{(1)}$ largely disappears. The measured NLL at this scale is -31.0363 (Table IV).



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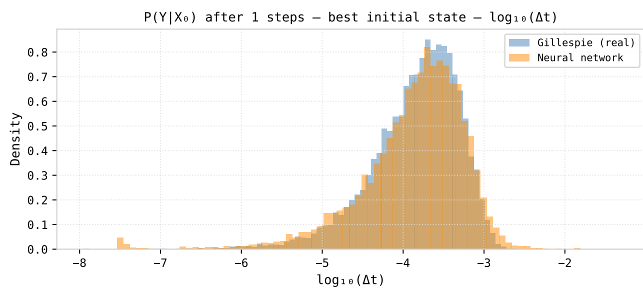


(b)

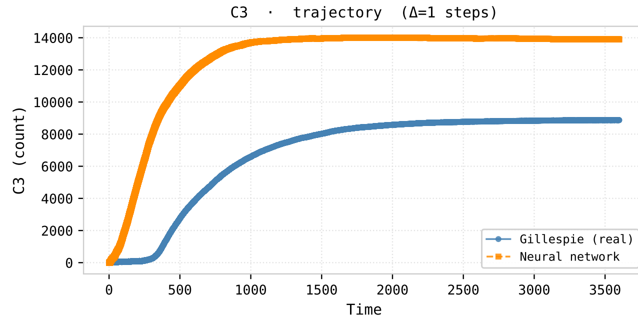
FIG. 4: Single-step distributions at scale $\Delta = 1$ for the initial condition with the lowest Wasserstein discrepancy. Each panel overlays the empirical SSA histogram (blue) against 10000 CNSF samples (orange). The Caspase-3 increment ΔN_{C3} is shown on a linear axis; the elapsed time Δt is shown on a $\log_{10}$ axis.

b. Single-Step Histograms. Figure 6 shows the single-step histograms at $\Delta = 100$. The marginal distribution of the Caspase-3 increment ΔN_{C3} is no longer concentrated on $\{-1, 0, +1\}$ but becomes bimodal, with one mode near 0 and a second mode close to 100. The $\Delta t^{(100)}$ distribution, being a sum of 100 exponential-like inter-event times, exhibits clear multimodality; nevertheless, the CNSF closely matches the reference histograms from the test set in both coordinates.

c. Wasserstein Analysis. Figure 7 and Table V present the Wasserstein results for this scale. Compared to $\Delta = 1$, the normalised distances W_1^{norm} do not increase substantially, indicating that the architecture continues to capture the per-step joint distribution well even after subsampling by a factor of 100. This suggests that the flow is able to model the mesoscopic transition kernel without a noticeable loss of fidelity.

Figure 8 illustrates the single-step distributions of ΔN_{C3} and Δt at $\Delta = 100$ for the *best* initial condition.

d. Autoregressive Trajectories. At $\Delta = 100$ the autoregressive chain exhibits improved long-horizon behaviour



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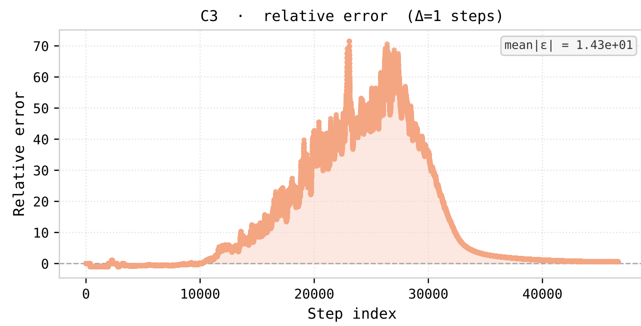

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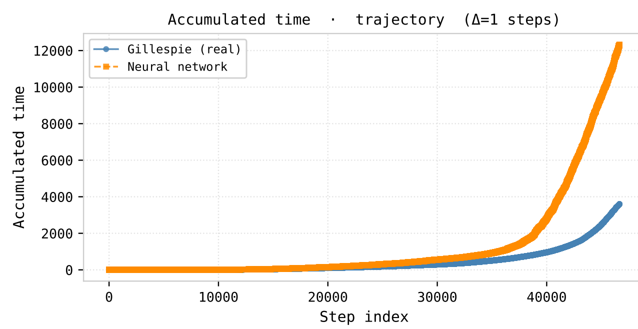

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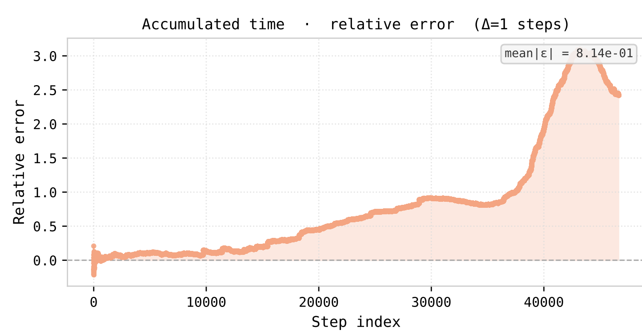

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FIG. 5: Autoregressive trajectory comparison at scale $\Delta = 1$ from the initial condition with the lowest Wasserstein discrepancy. *Top:* Caspase-3 counts (a) and accumulated time (c) for SSA (blue) and CNSF (orange, dashed). *Bottom:* corresponding relative errors $\varepsilon_{k,i}$ for Caspase-3 (b) and elapsed time (d); badges report the mean absolute relative error MARE_i.

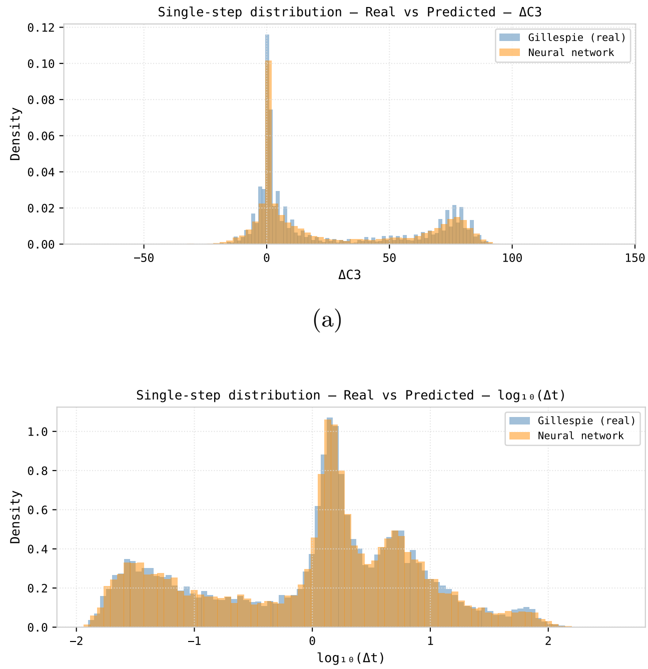

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FIG. 6: Single-step marginal distributions at scale $\Delta = 100$ for the Caspase-3 increment and the elapsed time. Each panel overlays the empirical SSA histogram (blue) against 10000 CNSF samples (orange). The Caspase-3 increment ΔN_{C3} is shown on a linear axis; the elapsed time Δt is shown on a $\log_{10}$ axis.

compared to $\Delta = 1$. To reproduce a trajectory over a fixed physical time span, the model now requires 10^2 times fewer steps: for example, a trajectory that would require 1000 one-step updates at $\Delta = 1$ can be represented with only 10 updates at $\Delta = 100$. This reduction in the number of autoregressive updates lowers the cumulative effect of per-step transition errors and yields trajectories that track the reference SSA dynamics more closely.

Figure 9 and Table VI present the trajectory comparison and the corresponding MARE values.

C. Scale $\Delta = 300$: Coarse-Grained Dynamics

a. Negative Log-Likelihood. At 300-step aggregation the target distribution captures macroscopic biochemical dynamics over timescales comparable to the transient induction phase of apoptosis. The learned model must encode complex, concentration-dependent drifts (driven by nonlinear propensity functions) alongside the increased variance induced by multi-step aggregation. The test NLL at this scale is -30.6531 (Table IV).

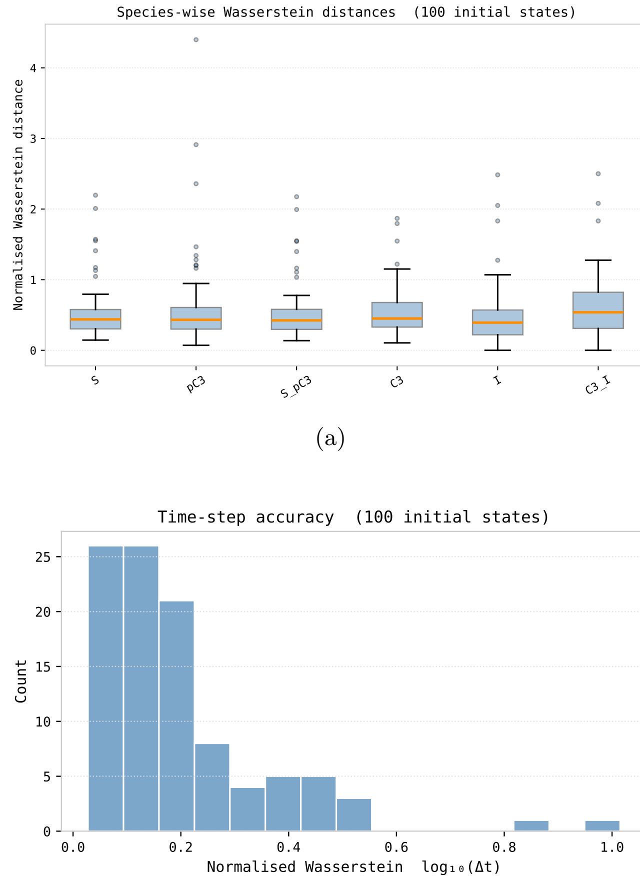

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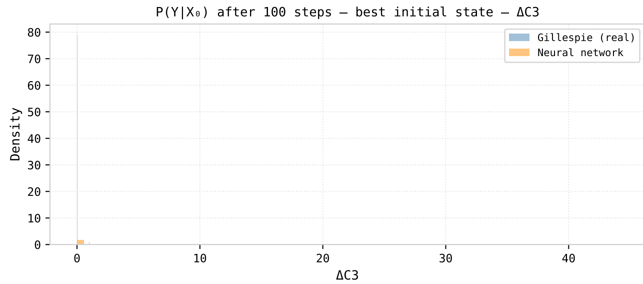
FIG. 7: Normalised Wasserstein-1 distances W_1^{norm} across 100 initial conditions at scale $\Delta = 100$. *Top:* Boxplots of W_1^{norm} for the single-step increments ΔN_i of each species; boxes span the interquartile range, whiskers extend to $1.5\times$ IQR. *Bottom:* Histogram of the 100 W_1^{norm} values for the time dimension, i.e. for the $\log_{10}(\Delta t)$ distributions.

b. Single-Step Histograms. Figure 10 shows the histograms at $\Delta = 300$. The bimodal and multimodal nature of the ΔN_{C3} and Δt distributions becomes more pronounced at this coarse resolution. Despite this increased complexity, the architecture is still able to infer output distributions that remain close to those produced by the Gillespie algorithm.

c. Wasserstein Analysis. Figure 11 and Table V present the Wasserstein results. As at finer scales, these results highlight the capacity of the model to predict single-step distributions accurately. It is noteworthy that for this subsampling factor the mean distances are generally lower, which is consistent with the single-step distributions becoming closer to Gaussian as the aggregation level increases.



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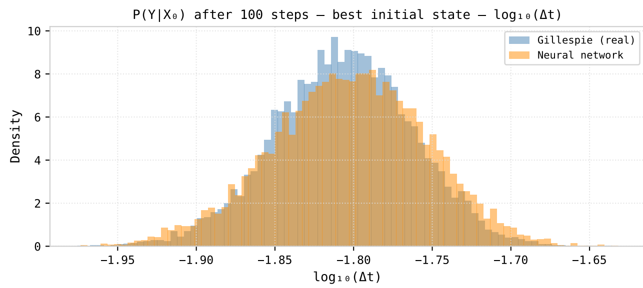

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FIG. 8: Single-step distributions at scale $\Delta = 100$ for the initial condition with the lowest Wasserstein discrepancy. Each panel overlays the empirical SSA histogram (blue) against 10000 CNSF samples (orange). The Caspase-3 increment ΔN_{C3} is shown on a linear axis; the elapsed time Δt is shown on a $\log_{10}$ axis.

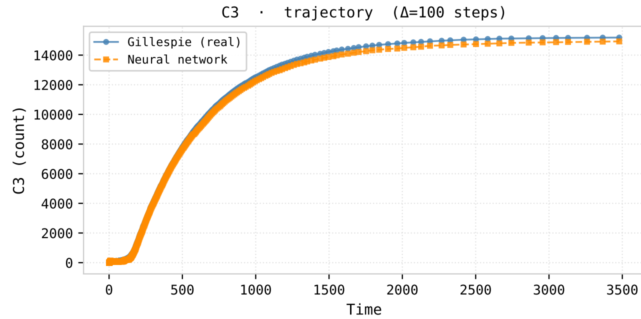
Figure 12 shows the single-step distributions of ΔN_{C3} and Δt at $\Delta = 300$ for the *best* initial condition, again defined as the one with the lowest total W_1^{norm}. As expected from the central limit theorem, the single-step distributions for a subsampling factor of 300 steps are close to Gaussian.

d. Autoregressive Trajectories. At the coarsest resolution, the number of steps in an autoregressive rollout is smallest, which further limits error accumulation per rollout, as already observed for $\Delta = 100$. Figure 13 shows that the inferred trajectories remain close to those simulated by the Gillespie algorithm over the full time horizon.

D. Cross-Scale Comparison

Tables V and VI consolidate the quantitative metrics across all three scales, enabling a cross-scale analysis of the trade-off between per-step distributional fidelity and long-horizon trajectory accuracy.

The cross-scale analysis reveals a nuanced picture. From a *per-step distributional* perspective (Table V), increasing the aggregation scale generally improves the



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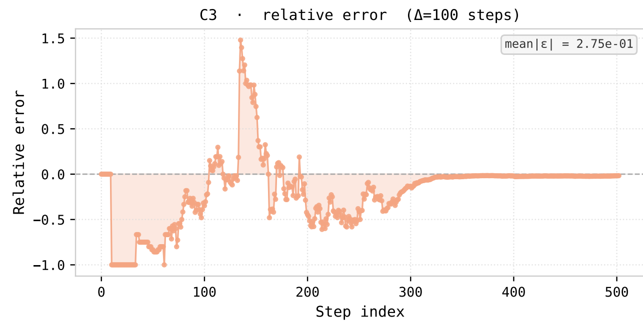

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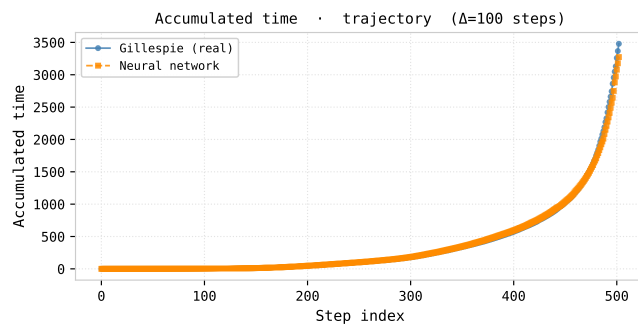

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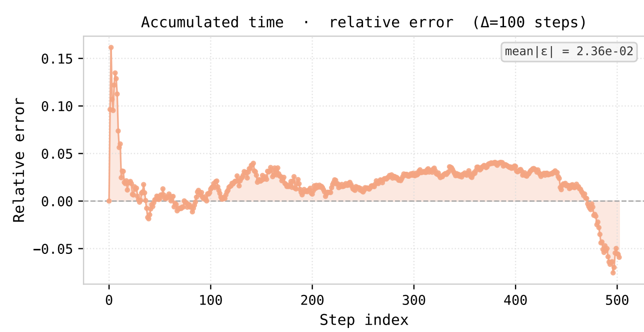

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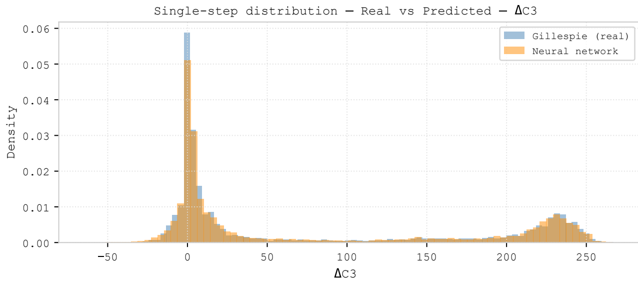
FIG. 9: Autoregressive trajectory comparison at scale $\Delta = 1$ from the initial condition with the lowest Wasserstein discrepancy. *Top:* Caspase-3 counts (a) and accumulated time (c) for SSA (blue) and CNSF (orange, dashed). *Bottom:* corresponding relative errors $\varepsilon_{k,i}$ for Caspase-3 (b) and elapsed time (d); badges report the mean absolute relative error MARE_i.



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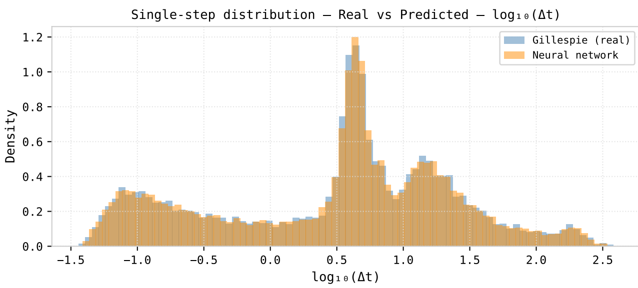

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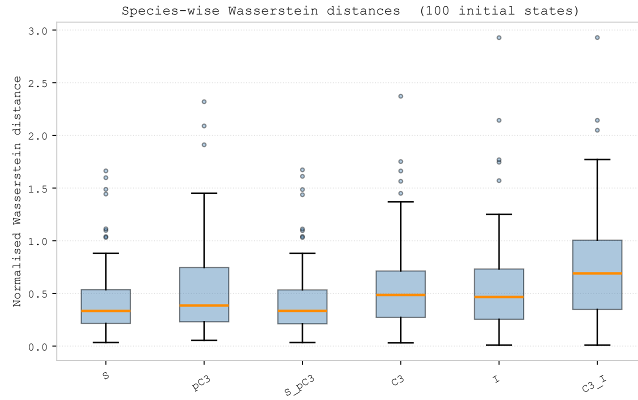
FIG. 10: Single-step marginal distributions at scale $\Delta = 300$ for the Caspase-3 increment and the elapsed time. Each panel overlays the empirical SSA histogram (blue) against 10000 CNSF samples (orange). The Caspase-3 increment ΔN_{C3} is shown on a linear axis; the elapsed time Δt is shown on a $\log_{10}$ axis.



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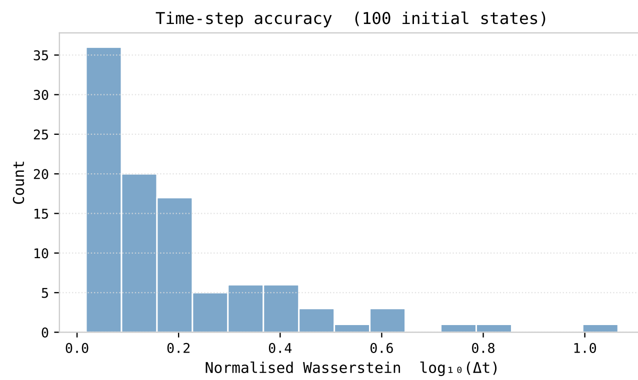

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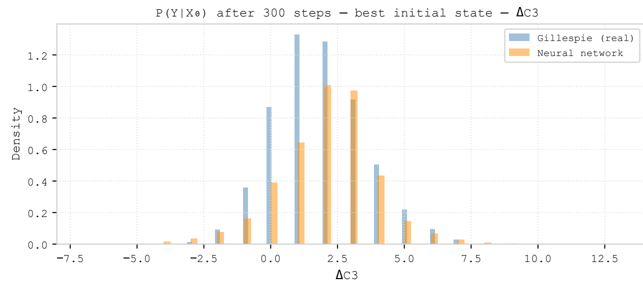
FIG. 11: Normalised Wasserstein-1 distances W_1^{norm} across 100 initial conditions at scale $\Delta = 300$. *Top:* Boxplots of W_1^{norm} for the single-step increments ΔN_i of each species; boxes span the interquartile range, whiskers extend to 1.5× IQR. *Bottom:* Histogram of the 100 W_1^{norm} values for the time dimension, i.e. for the $\log_{10}(\Delta t)$ distributions.

match between the CNSF and SSA kernels for the molecular species: mean W_1^{norm} values for all six species decrease markedly when moving from $\Delta = 1$ to $\Delta = 300$. This behaviour is consistent with the single-step increment distributions becoming smoother and closer to Gaussian under temporal aggregation. The time coordinate behaves differently: its Wasserstein distance remains small and stable between $\Delta = 1$ and $\Delta = 100$, but increases substantially at $\Delta = 300$, indicating that coarse time increments are harder to match faithfully than coarse molecular increments.

In contrast, the *trajectory-level* errors (Table VI) show a strong benefit from temporal aggregation for most coordinates. For S, S·pC3, C3, C3·I, and the accumulated time, MARE decreases by one to several orders of magnitude when moving from $\Delta = 1$ to $\Delta = 100$ or $\Delta = 300$, reflecting the reduced number of autoregressive updates and hence reduced error accumulation. However, this trend is not uniform: for pC3 and I the errors become large at $\Delta = 100$ and remain substantial at $\Delta = 300$, suggesting that these coordinates are particularly sensitive to coarse-graining, likely because their dynamics involve sharp transitions around the apoptotic threshold.

Overall, the data indicate that no single scale dominates across all metrics. Fine-grained sampling ($\Delta = 1$) offers the most faithful local transition kernel, especially for time, but suffers from strong error accumulation in long autoregressive rollouts. Coarse-grained sampling ($\Delta = 300$) greatly reduces trajectory errors for several species but degrades the fidelity of the time increments and certain sensitive species. A surrogate model intended for downstream inference tasks should therefore be selected by jointly considering the Wasserstein distances (per-step distributional accuracy) and the MARE values (trajectory-level fidelity), with the optimal scale depending on which coordinates and time horizons are most critical for the target application.



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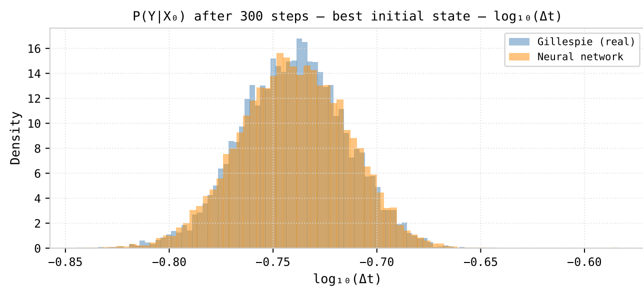

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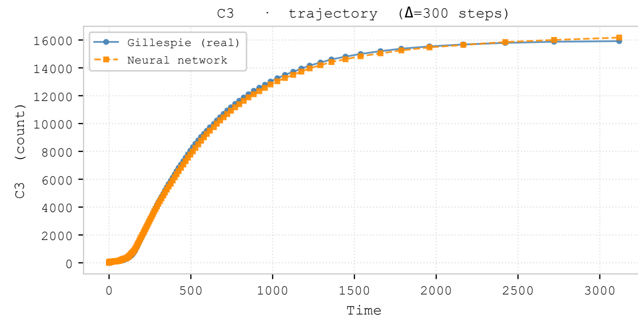
FIG. 12: Single-step distributions at scale $\Delta = 300$ for the initial condition with the lowest Wasserstein discrepancy. Each panel overlays the empirical SSA histogram (blue) against 10000 CNSF samples (orange). The Caspase-3 increment ΔN_{C3} is shown on a linear axis; the elapsed time Δt is shown on a $\log_{10}$ axis.



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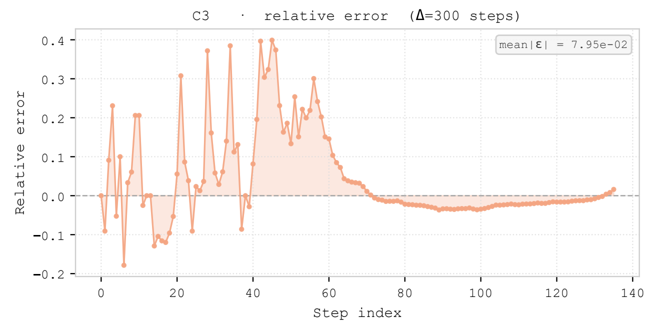

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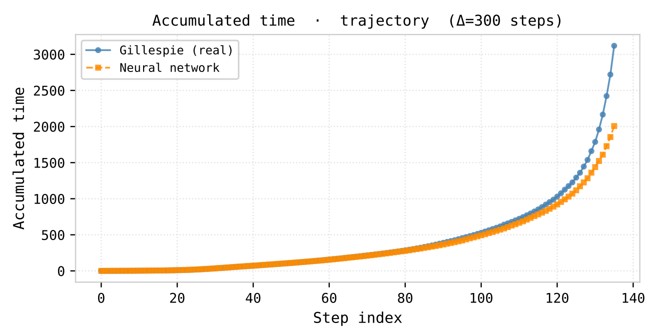

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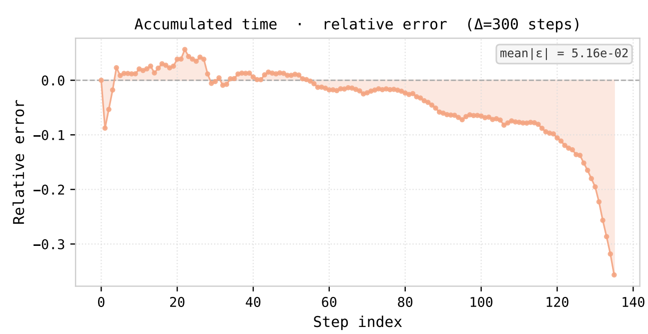

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FIG. 13: Autoregressive trajectory comparison at scale $\Delta = 1$ from the initial condition with the lowest Wasserstein discrepancy. *Top:* Caspase-3 counts (a) and accumulated time (c) for SSA (blue) and CNSF (orange, dashed). *Bottom:* corresponding relative errors $\varepsilon_{k,i}$ for Caspase-3 (b) and elapsed time (d); badges report the mean absolute relative error MARE_i.

VI. CONCLUSIONS AND PERSPECTIVES

We have presented and evaluated a Conditional Neural Spline Flow (CNSF) as a surrogate emulator for the stochastic transition kernel of a reduced Caspase-3 activation network. By learning the joint distribution $P(\Delta\mathbf{N}^{(K)}, \Delta t^{(K)} \mid \mathbf{N}_0)$ at three temporal resolutions ($K \in \{1, 100, 300\}$), the surrogate provides an exact, tractable likelihood for each multi-step transition and can be used as a drop-in replacement for the SSA at a chosen effective time resolution.

a. Main findings. Our multi-scale evaluation yields three main conclusions:

1. **Accurate single-step transition kernels across scales.** The CNSF attains test-set NLLs of -61.01, -31.04, and -30.65 at $\Delta \in \{1, 100, 300\}$, respectively (Table IV), consistent with the increasing complexity and broadening of the aggregated targets. Normalised Wasserstein distances (Table V) show that, for all six molecular species, the per-step discrepancy between the CNSF and SSA transition kernels *decreases* with temporal

TABLE V: Summary of normalised Wasserstein-1 distances (W_1^{norm}) across species and time for each temporal scale. Values are mean ± standard deviation over 100 initial conditions; the minimum over the same set is also reported. Lower is better.

(a) $\Delta = 1$

Species	Mean	Std	Min
S	0.5105	0.3518	0.1446
pC3	0.5651	0.5667	0.0705
S·pC3	0.5020	0.3499	0.1370
C3	0.5362	0.3227	0.1052
I	0.4584	0.3869	0.0000
C3·I	0.6050	0.4131	0.0000
$\log_{10}(\Delta t)$	0.1957	0.1624	0.0278

(b) $\Delta = 100$

Species	Mean	Std	Min
S	0.4317	0.3357	0.0342
pC3	0.5363	0.4417	0.0544
S·pC3	0.4319	0.3361	0.0340
C3	0.5417	0.4146	0.0310
I	0.5599	0.4536	0.0095
C3·I	0.7266	0.5073	0.0095
$\log_{10}(\Delta t)$	0.1943	0.1907	0.0177

(c) $\Delta = 300$

Species	Mean	Std	Min
S	0.1913	0.1806	0.0100
pC3	0.1511	0.1493	0.0043
S·pC3	0.2033	0.1991	0.0061
C3	0.3200	0.8683	0.0041
I	0.1417	0.2194	0.0004
C3·I	0.2156	0.3423	0.0001
$\log_{10}(\Delta t)$	0.4322	0.2848	0.0871

TABLE VI: Mean absolute relative error (MARE) per species and for accumulated time across all three temporal scales. Each value is computed from a single autoregressive rollout starting from the optimal initial condition (lowest total W_1^{norm}) for each scale.

Species / Metric	$\Delta = 1$	$\Delta = 100$	$\Delta = 300$
S	5.61×10^{-1}	1.27×10^{-2}	1.38×10^{-1}
pC3	7.38×10^{-1}	1.99×10^{1}	3.85×10^{-1}
S·pC3	1.05×10^{0}	1.81×10^{-2}	2.67×10^{-1}
C3	1.43×10^{1}	2.75×10^{-1}	7.95×10^{-2}
I	8.04×10^{2}	2.38×10^{0}	3.12×10^{1}
C3·I	4.73×10^{-1}	1.32×10^{-1}	3.26×10^{-2}
Acc. time	8.14×10^{-1}	2.36×10^{-2}	5.16×10^{-2}

aggregation: mean W_1^{norm} values drop from ≈ 0.5–0.7 at $\Delta = 1$ to ≈ 0.15–0.32 at $\Delta = 300$.

This is consistent with the single-step increments becoming smoother and closer to Gaussian under aggregation. In contrast, the time coordinate remains well modelled at $\Delta = 1$ and $\Delta = 100$ (mean $W_1^{\mathrm{norm}} \approx 0.20$) but becomes more challenging at $\Delta = 300$ (mean $W_1^{\mathrm{norm}} \approx 0.43$), indicating that coarse time increments are harder to reproduce faithfully than coarse molecular increments.

2. **Temporal aggregation mitigates error accumulation for most coordinates.** Autoregressive trajectory comparisons and the corresponding MARE values (Table VI) reveal a strong benefit from temporal aggregation for several key observables. For S, S·pC3, C3, C3·I, and the accumulated time, MARE decreases by one to three orders of magnitude when moving from $\Delta = 1$ to $\Delta = 100$, and remains small at $\Delta = 300$. For example, the MARE for S drops from 5.61×10^{-1} at $\Delta = 1$ to 1.27×10^{-2} at $\Delta = 100$, and the MARE on accumulated time decreases from 8.14×10^{-1} to 2.36×10^{-2} over the same change in scale. This behaviour reflects the reduced number of autoregressive updates required to span a fixed physical time interval at larger Δ, which limits the accumulation of small per-step errors.

3. **No single temporal scale is uniformly optimal.** The improvements brought by aggregation are not uniform across species. In particular, pC3 and I exhibit very large trajectory errors at some scales: $\mathrm{MARE}_{\mathrm{pC3}}$ grows to 1.99×10^{1} at $\Delta = 100$, while $\mathrm{MARE}_{\mathrm{I}}$ is 8.04×10^{2} at $\Delta = 1$ and remains sizeable at $\Delta = 300$ (3.12×10^{1}). These coordinates are strongly involved in the switch-like onset of apoptosis and appear especially sensitive to coarse-graining. Overall, fine-grained sampling ($\Delta = 1$) provides the most faithful local transition kernel (especially for time) but suffers from pronounced long-horizon error accumulation, whereas coarse sampling ($\Delta = 300$) yields smoother trajectories for several species but degrades the fidelity of time increments and certain sensitive species.

From a computational standpoint, it is important to emphasise that the current CNSF implementation is *not* faster than the Gillespie algorithm on a per-step basis. Each neural step involves multiple matrix multiplications and spline evaluations, whereas a single SSA reaction update only requires a handful of propensity evaluations and random draws. The practical advantage emerges only when the model is trained with a subsampling factor $K > 1$: a single CNSF call at $\Delta = 300$ advances the system by the effective time span of roughly 300 Gillespie reactions, whereas SSA must simulate those 300 events explicitly. The empirical trends in NLL, W_1^{norm}, and MARE therefore suggest that, for appropriately chosen subsampling factors, CNSF-based surrogates could achieve substantial *effective* speedups in long-horizon simulations, even if individual steps remain more expensive than a single SSA reaction.

b. Limitations and perspectives. Several directions naturally extend this work:

1. **Physical constraints.** The present architecture does not enforce integer-valued state updates exactly; the round-and-clip post-processing in Eq. (10) is an approximation that may distort rare-event statistics. Future work could incorporate integer-valued normalising flows or discretisation-aware training objectives to guarantee $\Delta\mathbf{N} \in \mathbb{Z}^6$ and strict non-negativity of $\mathbf{N}$.

2. **Scaling to the full apoptosis network.** Extending the approach from a six-species toy model to realistic apoptosis networks with 40 or more species will require careful study of how flow depth, conditioning networks, and preprocessing should scale with state-space dimensionality. Structured architectures (e.g. exploiting reaction graph sparsity) may be needed to maintain tractability.

3. **Multi-fidelity and multi-scale surrogates.** Here we trained three separate models for $\Delta \in \{1, 100, 300\}$. A promising extension is to train a single multi-fidelity surrogate, conditioned on the subsampling factor Δ, enabling dynamic trade-offs between accuracy and computational cost at inference time and potentially allowing cross-scale consistency constraints.

4. **Integration with simulation-based inference.** The exact likelihood of the CNSF makes it a natural likelihood approximator for simulation-based Bayesian inference frameworks such as sequential neural posterior estimation (SNPE) (Cranmer *et al.*, 2020; Lueckmann *et al.*, 2021). Embedding the surrogate within such pipelines would enable calibrated inference of biochemical rate constants or initial-condition distributions from experimental measurements, while amortising the cost of SSA-level accuracy.

5. **Stochastic optimal control and design.** The differentiable, sampleable surrogate dynamics could be coupled to policy gradient or model-based reinforcement learning methods for optimal drug-dosing strategies that steer apoptotic pathways toward desired outcomes. Likewise, the surrogate could support high-throughput in-silico screening of perturbations (e.g. inhibitor levels, pro-drug dosing schedules) prior to laboratory validation.

REFERENCES

K. Bose, and P. Chaudhari (2019), *Unravelling Cancer Signaling Pathways: A Multidisciplinary Approach* (Springer, Singapore).

E. Cramer, A. Mitsos, and M. Dahmen (2024), "Least squares and marginal log-likelihood model predictive control using normalizing flows for stochastic chemical processes," arXiv preprint arXiv:2409.17632 .

K. Cranmer, J. Brehmer, and G. Louppe (2020), "The frontier of simulation-based inference," Proceedings of the National Academy of Sciences **117** (48), 30055–30062.

C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios (2019a), "Neural spline flows," in *Advances in Neural Information Processing Systems*, Vol. 32 (Curran Associates, Inc.).

C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios (2019b), "Neural spline flows," in *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32, pp. 7511–7522.

C. Durkan, A. Bekasov, I. Murray, and G. Papamakarios (2020), "nflows: normalizing flows in PyTorch," .

D. T. Gillespie (1976a), "A general method for numerically simulating the stochastic time evolution of coupled chemical reactions," Journal of Computational Physics **22** (4), 403–434.

D. T. Gillespie (1976b), "A general method for numerically simulating the stochastic time evolution of coupled chemical reactions," Journal of Computational Physics **22** (4), 403–434.

D. T. Gillespie (1977a), "Exact stochastic simulation of coupled chemical reactions," The Journal of Physical Chemistry **81** (25), 2340–2361.

D. T. Gillespie (1977b), "Exact stochastic simulation of coupled chemical reactions," The Journal of Physical Chemistry **81** (25), 2340–2361.

D. T. Gillespie (2007), "Stochastic simulation of chemical kinetics," Annual Review of Physical Chemistry **58** (1), 35–55.

M. Hamedi, M. van der Merwe, A. P. Oliveira, and V. Hatzimanikatis (2024), "Generative machine learning produces kinetic models that accurately characterize intracellular metabolic states," Nature Catalysis **7** (10), 1086–1098.

K. He, X. Zhang, S. Ren, and J. Sun (2016), "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, resNet architecture; basis for the ResidualNet coupling networks used in each flow layer.

S. K. Lam, A. Pitrou, and S. Seibert (2015), "Numba: A LLVM-based Python JIT compiler," in *Proceedings of the Second Workshop on the LLVM Compiler Infrastructure in HPC*, pp. 1–6.

J.-M. Lueckmann, J. Boelts, D. S. Greenberg, P. J. Gonçalves, and J. H. Macke (2021), "Benchmarking simulation-based inference," in *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, Vol. 130 (PMLR) pp. 343–351.

D. R. McIlwain, T. Berger, and T. W. Mak (2013), "Caspase functions in cell death and disease," Cold Spring Harbor Perspectives in Biology **5** (4), a008656.

G. Papamakarios, E. Nalisnick, D. J. Rezende, S. Mohamed, and B. Lakshminarayanan (2021), "Normalizing flows for probabilistic modeling and inference," Journal of Machine

Learning Research **22** (57), 1–64.

A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala (2019), "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 32.

S. Raychaudhuri, E. Willgohs, T.-N. Nguyen, E. M. Khan, and T. Goldkorn (2008), "Monte Carlo simulation of cell death signaling predicts large cell-to-cell stochastic fluctuations through the type 2 pathway of apoptosis," Biophysical Journal **95** (8), 3559–3562.

C. Villani (2009), *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften, Vol. 338 (Springer).

M. Yang, P. Wang, D. del Castillo-Negrete, Y. Cao, and G. Zhang (2024), "A pseudoreversible normalizing flow for stochastic dynamical systems with various initial distributions," SIAM Journal on Scientific Computing **46** (4), C508–C533.

Appendix A: Use of AI/LLM Tools

1. Tools used

Tools used in the preparation of this report:

- ChatGPT (GPT–4o, accessed via `chat.openai.com`, May–June 2026). Used as a large language model (LLM) for language editing, rephrasing, and occasionally drafting or restructuring technical text.
- ChatGPT Image Creator (accessed via `chat.openai.com`, May 2026). Used to generate the schematic diagram of the Conditional Neural Spline Flow (CNSF) architecture (Fig. 1).

No other AI or LLM tools (e.g. Copilot, Claude, Gemini) were used in the writing of this report.

2. Text contributions

LLM contributions are reported using the four levels defined in the course guidelines[1].

Section-level summary

Section	Level	Notes
Abstract	2	LLM improved structure and wording based on my draft ideas
1. Introduction	2	Paragraphs rephrased for clarity; all content and citations by me
2. Biological system and SSA model	2	I wrote the technical content; LLM smoothed prose
3. CNSF surrogate model	3	LLM drafted initial exposition given my architecture description; I revised
4. Evaluation framework	2	LLM helped reorganise and rephrase metric descriptions
5. Results and discussion	2	I wrote the quantitative findings; LLM assisted with polishing and coherence
6. Conclusions and perspectives	3	LLM produced a first draft from my bullet points; I edited and checked details
Figures captions	2	LLM suggested shorter, clearer captions; I checked consistency with figures
Bibliography text (not entries)	1	Minor grammar and phrasing corrections

Notes on Level 3 sections

For the sections marked Level 3 (notably Sec. III and Sec. VI) the interaction pattern was as follows:

- For Sec. III (Conditional Neural Spline Flow surrogate), I provided ChatGPT with a description of the architecture and training setup (encoder, spline flow, preprocessing, and training hyperparameters) and asked for a coherent methods-style write-up suitable for a physics/ML report. The model produced an initial draft, which I then edited to match my notation, removed or corrected generic phrasing, and ensured all equations and algorithmic details agreed with my actual implementation and experiments.
- For Sec. VI, I gave ChatGPT a bullet-point summary of the main empirical findings (NLL values, Wasserstein trends across K, trajectory errors, and the inference-time considerations) and asked for a concise conclusions section. I then restructured the text, updated numerical references to match Tables IV–VI, and added clarifications about the fact that the per-step CNSF inference is not cheaper than a single Gillespie reaction but may be advantageous when trained with a subsampling factor $K > 1$.

In all cases, I read and understood the LLM output, corrected inaccuracies, and verified that every equation, claim, and numerical statement is consistent with the rest of the report and with my own simulations.

[1] Level 0: no LLM assistance; Level 1: language only (spell-check, minor phrasing); Level 2: editorial (rewriting/restructuring text based on my ideas); Level 3: generative (LLM drafted substantial text that I then reviewed, corrected, and integrated). See the official guideline for full definitions.

3. General statement

Throughout May and June 2026, I used ChatGPT primarily as a writing and editing assistant: to improve grammar, clarity, and structure; to suggest alternative phrasings; and, in a few cases, to draft initial versions of explanatory text from my prompts. All scientific ideas, modelling choices, parameter values, simulation results, and interpretations are my own. I have checked that no results, references, or numerical values were invented by the LLM and that I can explain every equation, figure, and claim in this report in my own words.